

# ROSHAN KUMAR

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## OBJECTIVE

Backend Engineer with practical experience designing and deploying microservices-based distributed systems using Node.js, PostgreSQL, MongoDB, and Apache Kafka. Experienced in event-driven architecture, JWT authentication, REST API design, Kubernetes orchestration, and GitOps CI/CD with ArgoCD. Looking to contribute to scalable backend platforms and high-reliability services.

## EXPERIENCE

### Assistant System Engineer

Tata Consultancy Services (TCS)

Oct 2024 – Present

Kolkata, India

- Completed structured training in **Java** and **Spring Boot**, covering enterprise REST API design, OOP principles, dependency injection, and SQL-backed persistence.
- Tech:** Java · Spring Boot · SQL · JavaScript

## EDUCATION

### B.Tech in Computer Science & Engineering

Vinoba Bhave University, Hazaribagh

2021 – 2024

## SKILLS

### Languages

Java, JavaScript, TypeScript

### Backend

Node.js, Express.js, Spring Boot

### Databases

PostgreSQL, MongoDB, MySQL

### Messaging

Apache Kafka

### Infra

Docker, Kubernetes, AWS EKS, ArgoCD

### Tools

Git, NGINX, JWT, Postman

## PROJECTS

- Pizza Delivery Platform** [mern-pizza-app.cyou](https://mern-pizza-app.cyou)  
Production-grade, multi-tenant pizza ordering platform built across **5 independently deployed Node.js microservices**.
  - Architected **role-based authentication** (Admin / Manager / Customer) in a dedicated Auth Service backed by **PostgreSQL**; implemented JWT access + refresh token rotation with HTTP-only cookies; downstream services validate tokens locally via shared secret, removing Auth Service as a synchronous dependency and eliminating it as a single point of failure.
  - Built an **event-driven product price cache** in Order Service by consuming **Apache Kafka** events (PRODUCT\_UPDATED, TOPPING\_UPDATED) published by Catalog Service — removes cross-service HTTP calls at checkout and ensures price validation remains available even during Catalog Service downtime.
  - Integrated **Stripe PaymentIntent** API with webhook signature verification (POST /payments/webhook); order status transitions to PAID only after cryptographic confirmation from Stripe, preventing duplicate charges on network retries.
  - Designed a 9-event Kafka topology across 3 topics (catalog, orders, payments) enabling fan-out: a single ORDER\_PLACED event is independently consumed by WebSocket Service (real-time UI push) and Notification Service (email delivery) with no coupling between consumers.
  - Deployed on **AWS EKS** managed by **ArgoCD v3.3.2** (GitOps); GitHub Actions builds Docker images on push, updates K8s manifests; NGINX Ingress handles SSL termination, CORS headers, and path-based routing; 3 replicas for Order Service for peak-load tolerance.
  - Tech:** Node.js · PostgreSQL · MongoDB · Apache Kafka · Stripe · Socket.io · AWS S3 · Kubernetes · Docker · ArgoCD · NGINX · Next.js · React

[github.com/roshan798/pizza-delivery-app](https://github.com/roshan798/pizza-delivery-app)

- **VoiceHub** — Real-time Group Voice Calling App  
Built a real-time communication platform using **WebRTC** for peer-to-peer audio and **Socket.IO** for signalling and room lifecycle; REST API with **Node.js** + **Express** and **MongoDB** persistence.  
[github.com/roshan798/voicehub.me](https://github.com/roshan798/voicehub.me)

## ACHIEVEMENTS

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- Achieved **1750+ contest rating** on LeetCode; participated in **56+ contests** and ranked in the **global top 1300** in Weekly Contest 376.
- Solved **600+ problems** across [LeetCode](#), [GeeksForGeeks](#), and [HackerRank](#).
- Earned **6-star Gold** badge in HackerRank Problem Solving.

## CERTIFICATES

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- [HackerRank Problem Solving](#)
- [freeCodeCamp Responsive Web Design](#)
- [Hacktoberfest Contributor](#)